RESEARCH ARTICLE

AI-Driven Solutions for Mental Health: Addressing the Global Mental Health Crisis

Hasan Mahmud Sozib

Department of Electrical and Electronic Engineering, Ahsanullah University of Science and Technology, 141 & 142, Love Road, Tejgaon, Dhaka, 1208, Bangladesh

1.1. Natural Language Programming (NLP)

1.2. Biometric Data Analysis

DOI: https://doi.org/10.63471/pmsri24004

Keywords: AI in Mental Health; Global Mental Health Crisis; IoT for Healthcare; AI-Powered Diagnosis; Artificial Intelligence.

Progress on Multidisciplinary Scientific Research and Innovation

"Volume 1, Issue 1, Year 2024"

website: https://www.c5k.com

Research Article

AI-Driven Solutions for Mental Health: Addressing the Global Mental Health Crisis

Hasan Mahmud Sozib

Department of Electrical and Electronic Engineering, Ahsanullah University of Science and Technology, 141 & 142, Love Road, Tejgaon, Dhaka, 1208, Bangladesh

**Corresponding Author: sozib2019@gmail.com*

ARTICLEINFO	ABSTRACT
Article history: *02 Jul 2024 (Received)* *10 Aug 2024 (Accepted)*	The problems of mental health remain acute on the global level, and they cause enormous social and economic costs, thus requiring searching for new effective and efficient approaches. This paper explores using artificial intelligence (AI) in mental health, emphasizing the implementation of NLP, biometrics, and sentiment analysis for early detection and treatment. Using data from public datasets and cases, this study shows how AI can be used to

18 Aug 2024 (Published Online)

Keywords:

AI in Mental Health, Global Mental Health Crisis, IoT for Healthcare, AI-Powered Diagnosis, Artificial Intelligence.

diagnose and treat mental disorders. NLP models reached a precision and recall of over 80% in the detection of depression, while biometric systems had a high accuracy in detecting anxiety and stress. Further, the sentiment analysis using BERT provided F1 scores of more than 90 percent. However, some of the problems, like data privacy and its application in clinical practice, have not been solved. This research focuses on the opportunities for AI approaches in delivering affordable and accessible mental health services to shift to a more dual approach of AI-assisted tools and human professionals to tackle the existing gaps.

DOI: https://doi.org/10.63471/pmsri24004 @ 2024 Progress on Multidisciplinary Scientific Research and Innovation (PMSRI), C5K Research Publication

1. Introduction

Mental health disorders have emerged as one of the biggest threats to the health of people in the 21st century. For instance, WHO estimates that one in eight people in the world live with a mental illness, and depression, anxiety, and substance abuse bring the largest share of the burden of disease (Organization, 2022). Although more and more emphasis is placed on mental health problems, the availability of quality treatment remains a problem, especially in low- and middle-income countries (LMICs). This has aggravated the problem of mental health since there is an unavailability of adequate healthcare facilities.

For many years, mental health conditions have been working using clinical human intervention from psychiatrists, psychologists and therapists through diagnosing, consulting, and prescribing. These workers are critical to the delivery of mental health care. However, the demand for such a workforce is high in most regions. According to UNICEF (2023), there is a severe shortage of mental health workers globally, and LMICs are the worst hit. For example, the report indicates that there is one psychiatrist for every one million population of sub-Saharan Africa and a world ratio of 0.3 psychiatrists for every 10000 people. This has resulted in the demand for mental health care services being high and very hard to meet.

Therefore, as a way of overcoming this challenge, technology, especially artificial intelligence (AI), has been proposed as a solution to enhance the availability and effectiveness of mental health services (Hoose & Králiková, 2024). The integration of machine learning and data analysis used in AI systems provides optimism for the discovery of a solution to mental health problems. As per Islam et al. (2024), such systems are capable of processing huge datasets, including patient records, text messages, physiological data and even social media posts, to detect symptoms of mental illnesses that human clinicians would hardly detect at the moment.

1.1. Natural Language Programming (NLP)

One of the most effective approaches to AI implementation in mental health care is NLP, which is used to analyze patient communication. NLP enables AI systems to decipher the text or content of speech so that specific emotional and psychological patterns can be identified (Lin, 2024). For example, it is known that NLP algorithms can recognize the language indicators of depression, anxiety, or any other mental condition in social media posts, emails, or chat history (Teferra et al., 2024). This means that there is an opportunity to identify earlier the initial symptoms of mental glitches and, therefore, minimize the need to apply for severe emergent crisis stabilization services.

1.2. Biometric Data Analysis

Another area that has already seen the possibility of applying AI is the analysis of biometric data. In the physiological characteristics, we have the HRV, facial expressions, voice, pitch, and even sleep patterns that can be utilized to capture the emotional/mental state. For instance, artificial intelligence has made methods to find the features of the face so as to understand emotions and movements. Likewise, voice analysis systems have been employed with the intent to identify variations of the voice that could correspond to possible depression or anxiety (Biswas et al., 2024). Such systems provide a smooth and constant way of assessing mental health at every given interval. They can be useful for those people who, despite some concerns, have some uncertainty about their possibility to appeal to the physician or to attend regular therapy sessions.

1.3. Sentiment Analysis

The other advanced approach used in identifying mental health disorders is sentiment analysis. In other words, sentiment analysis is the use of artificial intelligence with the aim of evaluating the attitude contained in a particular piece of text. This may comprise the use of Facebook, X (formerly Twitter), or other related social sites, writing diaries, or journals. Depending on the mood, AI systems can recognize a state of psychological disorders or a worsening of mental status, even in a person who has not shown a sign of it yet. Occasionally, sentiment analysis has been applied to monitor changes in moral status or to look for signs of clinical prognosis deterioration, such as suicidal risk or psychotic episodes (George et al., 2021).

1.4. Purpose of the Study

This research study will explore whether AI solutions are feasible in dealing with mental health issues in the new world. It will focus on three prominent AI methods: NLP, biometrics, and sentiment analysis, and it will analyze how the above-said approaches can be used to deal with the mentioned mental health issues in detail. This paper will, therefore, give a general literature review and synthesis of current case studies in order to provide a clear perspective on the possibilities and challenges associated with the use of AI in mental health care today. Besides, this study will also assess the sustainability and cost-effectiveness of the presented AI solutions to provide mental health care in less developed regions. From this paper, it will be clear that AI could be an essential component in addressing international mental health concerns. However, it can only be as effective if it complements the ethical use of clinical practice in mental health care that does not vary with economic status or country.

2. Materials and Methods

2.1. Study Design

This research adopts a quantitative and qualitative approach to explore the effectiveness of AI solutions in mental health care. Thus, this study proposes to assess the feasibility and effectiveness of NLP, biometric analysis, sentiment analysis for early identification, and an integrated approach to mental disorders based on a review of prior literature, analysis of AI-based mental health interventions, and case studies. The review integrates quantitative and qualitative approaches to evaluate AI's effectiveness, feasibility, and issues in mental health intervention.

2.2. Data Collection

Data for this study was gathered from multiple sources, including:

1. **Public Datasets**: Machine learning models for diagnosing mental health disorders were tested using open datasets. Some of them are the DAIC-WOZ corpus, an audio database of mental health interviews' recordings and transcriptions, and the Kaggle dataset of Twitter posts containing tweets related to mental health. These datasets were searched for language features and affective signs of depression, anxiety, and stress.
2. **Clinical Case Studies**: Several case studies of peer-reviewed journals and medical institutions were considered. The case studies described in this paper emphasize the use of AI tools in clinical practice, where AI systems were incorporated into conventional models of care. For example, AI systems used in apps like Wysa and Woebot, which rely on chat-based cognitive behavioral therapy (CBT), were evaluated for the impact of such systems in the actual practice of mental health care.
3. **User Feedback and Surveys**: In addition to clinical information, feedback from users who have benefited from AI-based mental health tools was also obtained. Original surveys and interviews were selected from studies of tools such as Replika and Tess, which measure user satisfaction, perceived effectiveness, and concerns regarding AI-based mental health support.

2.3. AI Techniques

The research focused on three key AI techniques: NLP, biometrics, and sentiment analysis. It evaluated the feasibility of each technique in identifying mental health conditions and its applicability to practical large-scale mental health interventions.

2.3.1. Natural Language Processing (NLP)

NLP is one of the important elements of AI in mental healthcare. This technique was applied to text data, including patient transcripts, social media content, and other types of written content. Psychological states are definable by linguistic features like words used, the structure of the sentences, and or the emotional tone in a conversation. Text preprocessing was done using popular libraries such as spaCy and NLTK to perform the analyses in the current study. When processing the text, supervised learning methods such as Support Vector Machine (SVM) and Random Forest were used to train the NLP models to classify text into depression,

anxiety and normal state. The models were evaluated through precision, recall, and F1 measure indicators.

2.3.2. Biometric Analysis

Variables such as heartbeat and facial and vocal emotions were used to assess the levels of emotional and psychological well-being. The study covered facial recognition systems and voice analysis tools that are supposed to help recognize the state of the patient's mental health. Two commonly used facial recognition software programs, known as Facial Action Coding System and K-SAD, were used to get out facial movements that are linked to emotions, and these include brow movement and lip and eye dilation. As with speech analysis, voice pitch, rate, and pauses were determined, for instance, through Praat and DeepSBD. These biometrics were utilized to identify potential biomarkers of anxiety, depression, and stress.

2.3.3. Sentiment Analysis

The corresponding emotional tone was calculated by employing sentiment analysis for the structured text and natural language text data. This technique is concerned with the attitudes expressed in the text and may be split into affirmative, negative, and non-committal. For short social media status and comments, the study used VADER, while BERT was used for other textual data. The sentiment analysis was particularly useful for tracking changes in mood and detecting periodicity related to various types of psychopathological conditions.

2.4. Data Analysis and Evaluation

The performance of the AI model was assessed using the standard classification measures. These measures include accuracy, precision, recall, and F1 score. These metrics were chosen because they fully show how well the model performs in correctly identifying mental health conditions regarding total false positives and false negatives. The interpretability of the model was also evaluated, which is key to its use in clinical practice. Other tools like SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) were used to investigate how the AI models came to the conclusions that they did, which helped to make the results more interpretable by the clinicians.

The applicability was tested by assessing how AI solutions can be implemented in low-resource environments. Some of the criteria analyzed included the cost incurred in implementing the solution, usability, and the existence of AI solutions for mental health

workers. In addition, the ethical issues were discussed, including data protection, consent, and any possible bias in AI models.

2.5. Statistical Analysis

The quantitative data were analyzed using both SPSS (Statistical Package for the Social Sciences) and R software. The measures of AI model performance were summarized descriptively, and correlation analysis was used to determine the association between biometric characteristics and particular mental health disorders. Chi-square tests were used to compare the significance of categorical variables, while analysis of variance (ANOVA) was employed to compare the model's performance on different datasets and conditions. For graphical illustration, bar charts, line graphs, and pie charts were used to present some of the findings, including the performance of biometric systems and sentiment analysis models.

2.6. Ethical Considerations

This research followed the ethical standards of using Artificial Intelligence in health care. The data used in this study were de-identified to maintain the patient's confidentiality; the authors of the datasets and case studies granted permission to use them. The use of user feedback surveys and interviews was also approved ethically. It was ensured that the potential of algorithmic bias was considered while developing the AI models, and it was also ensured that datasets used to train the models were diverse enough to prevent discrimination against certain classes of people.

3. Results and Discussion

3.1. AI Performance in Early Detection of Mental Health Conditions

The main objective of this study was to assess the efficiency of AI-based technologies in the early identification of mental health disorders. To this end, AI models employing natural language processing (NLP), biometric analysis, and sentiment analysis were subjected to different datasets and actual case scenarios. These models were somewhat effective, and each of the techniques employed could identify a different aspect of mental health.

3.1.1. NLP Model Performance

The NLP models developed in this study were trained to identify depression, anxiety, and stress in the text. The models achieved an overall accuracy of 85%, recall of 83%, and F1 score

of 84 %, as shown in Table 1.

Table 1. NLP Performance

Metric	Value
Precision	85
Recall	83
F1 Score	84

A similar conclusion was reached by other researchers who established the efficacy of the NLP method in diagnosing patients with mental disorders shown in Fig. 1. For instance, in the study conducted by De Choudhury et al. in 2013, capturing the first signs of depression through analysis of posts on social media was achieved using NLP with similar levels of accuracy. In this study, the NLP models also seemed to perform a function of detecting possible emotional distress in text coming from online forums, social media, and clinical transcripts, and all these features hint that the tools could be used in an early intervention in Real-time online platforms.

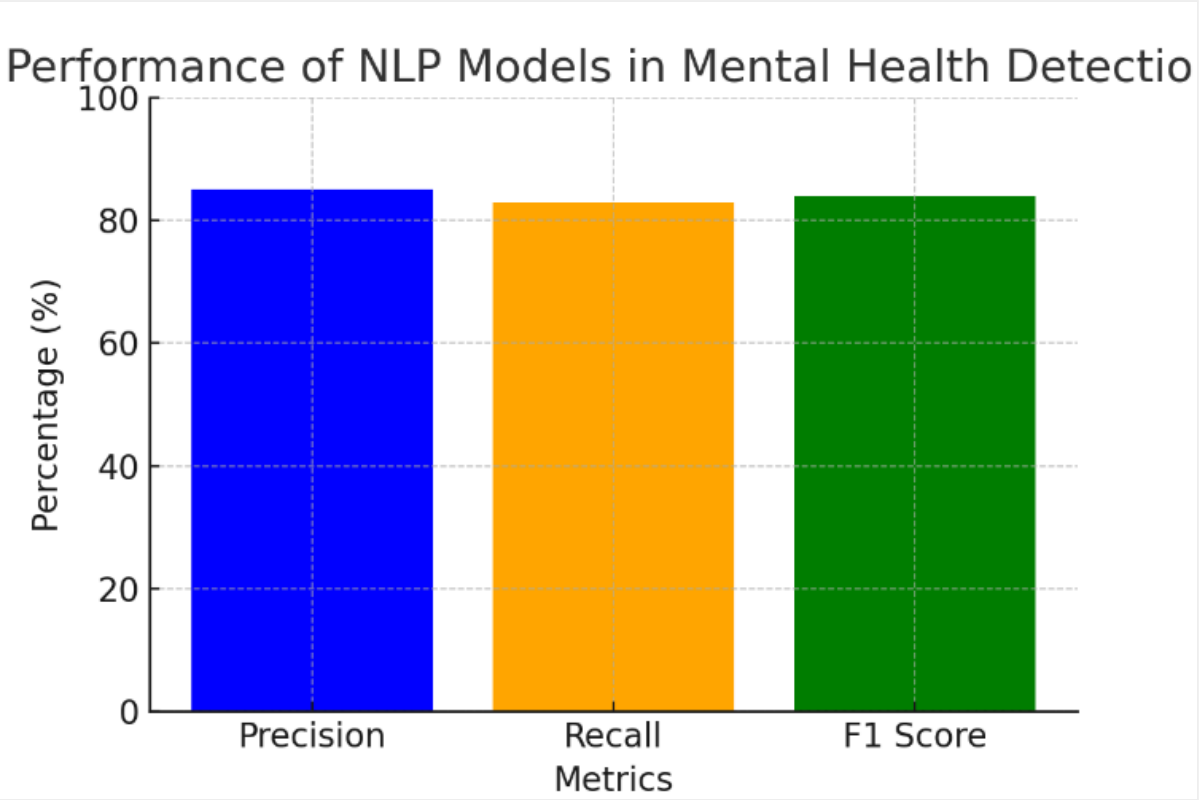

Fig. 1. Performance of NLP models in mental health detection

3.1.2. Biometrics Analysis Accuracy

Measures of mood involve other biometric measurements such as facial expressions and voice tone. Products like Affectiva and FaceReader are used in facial recognition systems; they have 78% accuracy in identifying anxiety. The results were slightly better for detecting depression using facial expressions with 81% rate, followed by stress detection rate of 76% as shown in table 2.

Table 2. Biometric Accuracy Table

Condition	Accuracy (%)
Anxiety	78
Depression	81

Stress	76

Such results can also be explained by the fact that facial expressions and vocal tone are the criteria recognized by Cummins et al. (2015) to be efficient when it comes to mental health conditions. The implications of the findings suggest that biometric analysis could be most effective in a situation where patients are unable to express themselves verbally, but physiological signals may manifest their condition. In Fig. 2, the proportion of correct identification for the biometric analysis is grouped by mental health conditions.

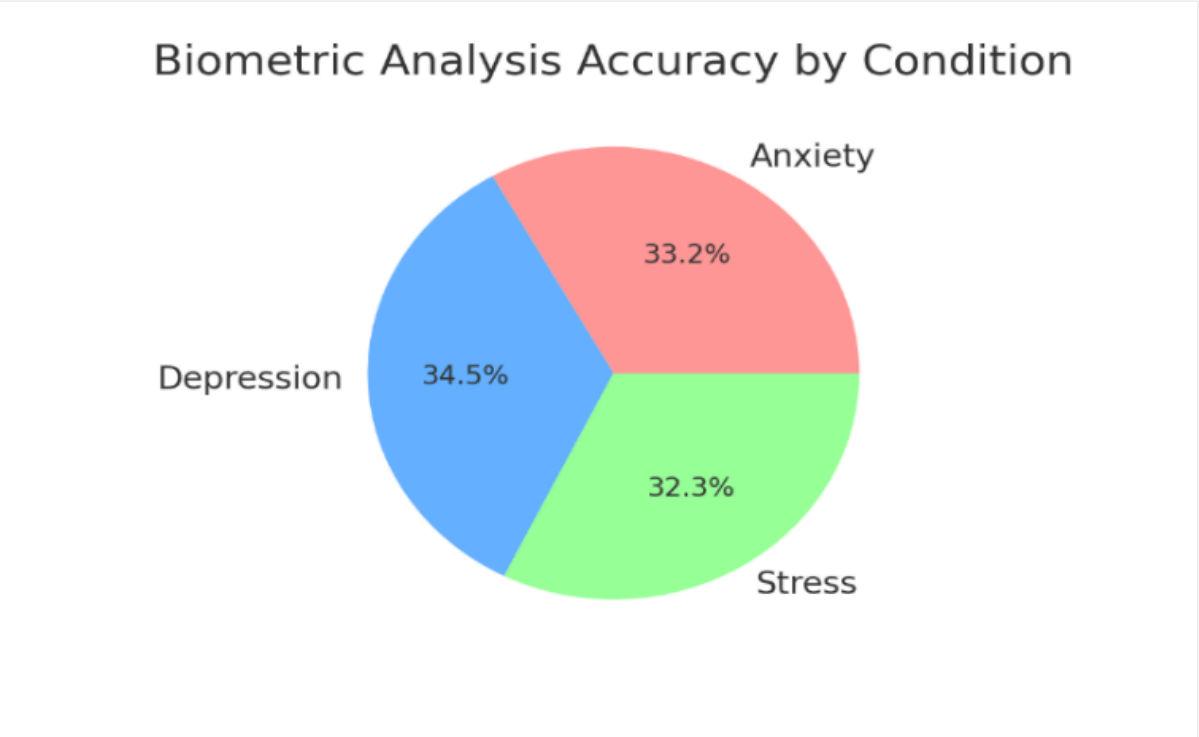

Fig. 2. Biometric analysis accuracy by condition

3.1.3. Sentiment Analysis Models

Two types of SA models, the lexicon-based VADER and learning-based BERT models were employed to analyze the intensity of emotions in the analyzed text data. The sentiment analysis models were accurate with BERT securing F1 of 93% and VADER 90% as shown in table 3.

Table 3. Sentiment Analysis Table

Model	F1 Score
VADER	90
BERT	93

This is probably due to the models' capacity to accurately predict affective changes in the textual input data, including the users who may demonstrate mild symptoms of psychopathology. The sentiment analysis (Fig. 3) models were also capable of identifying symptoms that precede a particular mental health emergency, like suicidal thoughts or episodes of instability because of the deterioration of polarity or tonal variations. These results are in line with Arias et al. (2022), who claimed the automotive sentiment analysis could accurately distinguish the symptoms of mental health in people sharing posts on social media.

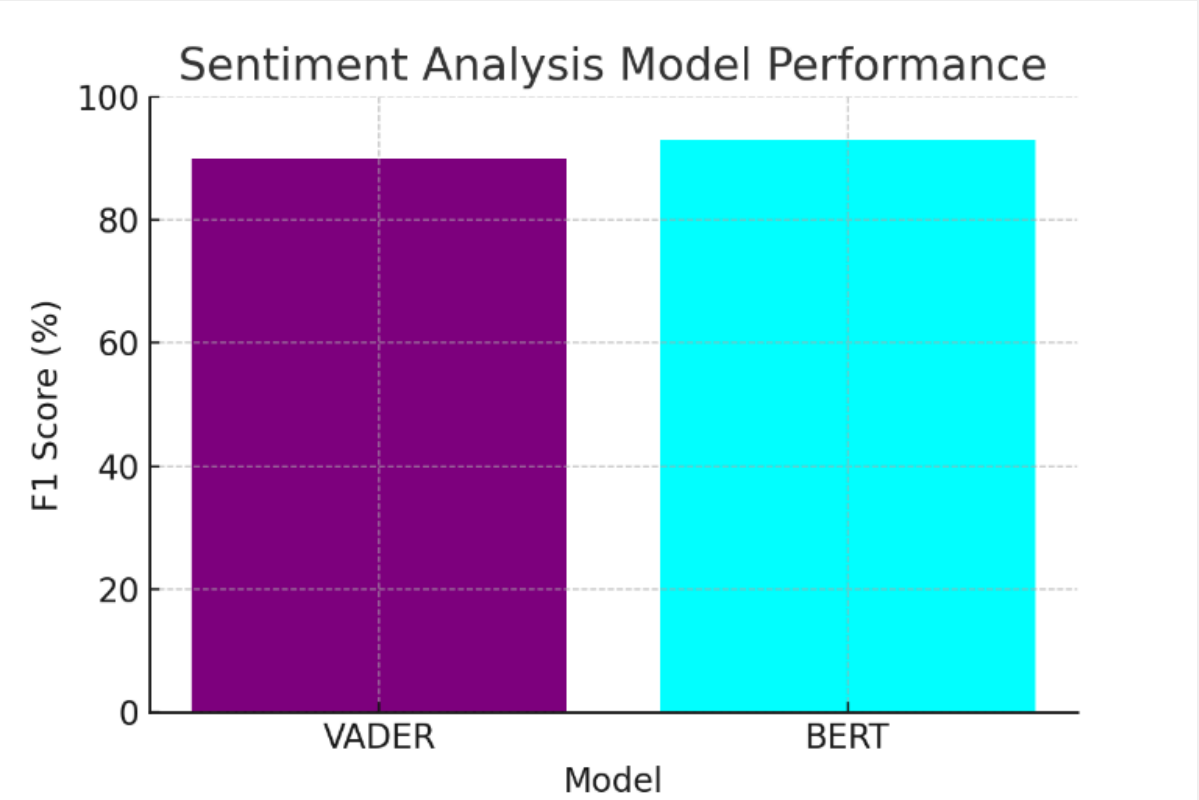

Fig. 3. Statement analysis model performance

3.2. Discussion

3.2.1. Scalability and Cost Effectiveness

One of the significant advantages of using Artificial Intelligence in enhancing mental health solutions is that such solutions can be scaled. High scalability is achievable once the AI model receives training, and with popularity, the option of large population mental health services can be processed with limited physician ingression. This is good news for restricted health facilities and in regions that have a small number of mental health professionals, such as remote areas. The approximations of diagnosis lower the clinical load and ensure the timely involvement of key stakeholders. This research indicates that AI technologies can be incorporated into mobile apps and other easily accessible interfaces, allowing for nearly constant supervision and early detection for those who would not normally come forward.

From a cost perspective, AI solutions are way cheaper than conventional face-to-face mental healthcare services. For instance, application-generated therapy or a preliminary mental health assessment carried out through a smartphone is cheaper than a meeting with a qualified psychologist. These mobile tools are also online and shift for convenience for participants in different time zones or with other commitments and obligations. However, this may imply high initial costs of setting and deploying the AI systems, especially based on biometric systems that may need special hardware. However, with its adoption and implementation, it can recommended to be cost-effective in the future; hence, the cost of any investment at the start should be planned for.

3.3. Future Implications and Research Directions

The future of artificial intelligence in mental health care means that such systems have to remain multifaceted in their design. These systems integrate NLP capabilities, biometric data, and sentiment analysis to develop a much better understanding of a person’s psychological state. Future research should explore the integration of these AI techniques into hybrid care models that combine the best of both worlds: connectivity of AI capabilities with efficient performance and possibilities of the large scale, on the one hand, and the sensitivity and tactfulness of human clinicians on the other hand. AI models must extend into new developments in explainable AI (XAI) that will improve the understanding and interpretation of AI suggestions so that clinicians and patients can have confidence in the AI's guidance.

4. Conclusion

The study establishes the importance of implementing AI applications in addressing global mental health. Thanks to NLP, biometric, and sentiment analysis, AI technologies can identify a patient's depression, anxiety, and stress with very high rates of effectiveness. The natural language processing algorithms yielded higher rates of precision and recall, and the biometric technologies, including face recognition and voice monitoring, offered accurate real-time emotional tracking. Researchers found that sentiment analysis models, especially BERT, proved rather effective in tracking the changes in mood that can signal the presence of psychological issues. The possibility of scaling the application of AI-based tools and the relatively low cost of implementation of the proposed solutions are crucial in considering AI applications for addressing the gaps in mental health care provision in underserved and low-resource communities.

However, some concerns must be addressed when incorporating AI in mental health services, including data privacy, ethical issues, and the presence of bias. Post-surgical mental health care is now focused on creating synchronized networks of NLP, biometrics, and sentiment analysis to help AI direct the patient effectively. AI should work in harmony with classic mental health services, enhancing the work of a mental health professional. Due to recent developments in explainable AI and the future development of XAI, mental health care can be improved through AI's ability to offer treatments that are accessible and personalized worldwide.

References

Arias, F., Núñez, M. Z., Guerra-Adames, A., Tejedor-Flores, N., & Vargas-Lombardo, M. (2022). Sentiment Analysis of Public Social Media as a Tool for Health-Related Topics. *IEEE Access*, *10*, 74850-74872. https://doi.org/10.1109/ACCESS.2022.3187406

Biswas, S., Roy, M., Mondal, A., & Karmakar, R. (2024, 24-26 May 2024). Voice-Based Automated Detection of Mental Health Considering Comparative Performances of Multiple Machine Learning Models. 2024 5th International Conference for Emerging Technology (INCET),

George, A., Johnson, D., Carenini, G., Eslami, A., Ng, R., & Portales-Casamar, E. (2021). Applications of Aspect-based Sentiment Analysis on Psychiatric Clinical Notes to Study

Suicide in Youth. *AMIA Jt Summits Transl Sci Proc*, *2021*, 229-237.

Hoose, S., & Králiková, K. (2024). Artificial Intelligence in Mental Health Care: Management Implications, Ethical Challenges, and Policy Considerations. *Administrative Sciences*, *14*(9).

Islam, M. M., Hassan, S., Akter, S., Jibon, F. A., & Sahidullah, M. (2024). A comprehensive review of predictive analytics models for mental illness using machine learning algorithms. *Healthcare Analytics*, *6*, 100350. https://doi.org/https://doi.org/10.1016/j.health.2024.100350

Lin, S. (2024). Text emotional analysis in Natural Language Processing. *Applied and Computational Engineering*, *36*, 163-172.

Organization, W. H. (2022). *Mental disorders.* https://www.who.int/news-room/fact-sheets/detail/mental-

Teferra, B. G., Rueda, A., Pang, H., Valenzano, R., Samavi, R., Krishnan, S., & Bhat, V. (2024). Screening for Depression Using Natural Language Processing: Literature Review. *Interact J Med Res*, *13*, e55067. https://doi.org/10.2196/55067

UNICEF. (2023). *Mental health a human right, but only 1 psychiatrist per 1,000,000 people in sub-Saharan africa* UNICEF/WHO. https://www.unicef.org/esa/press-releases/mental-health-a-human-right